

SINDH SOLAR ENERGY PROJECT (SSEP)

IMPLEMENTATION REVIEW & POLICY ASSESSMENT

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Section 1: Background and Policy

Pakistan's electricity crisis at the time of the Sindh Solar Energy Project was not simply a shortage of generation. The sector was also affected by unreliable supply, financial constraints, and institutional weaknesses. An Asian Development Bank evaluation described Pakistan as having experienced a severe energy crisis over the preceding decade, while noting that problems of circular debt and financial sustainability remained unresolved despite improvements in generation and transmission.¹ Electricity access remained particularly limited in rural and remote areas. Pakistan's National Action Plan: Sustainable Energy for All reported national electricity access at 73% and identified 32,266 villages likely to remain without grid access due to their remote locations and dispersed populations.² Sindh had the highest number of un-electrified villages among the provinces. This was also a period when renewable energy was becoming increasingly significant in Pakistan's electricity planning. The World Bank's assessment of the country's power system identified solar and wind as increasingly competitive sources of new generation.³ The SSEP was consequently designed to address both sides of the problem: increasing renewable generation and extending electricity access.

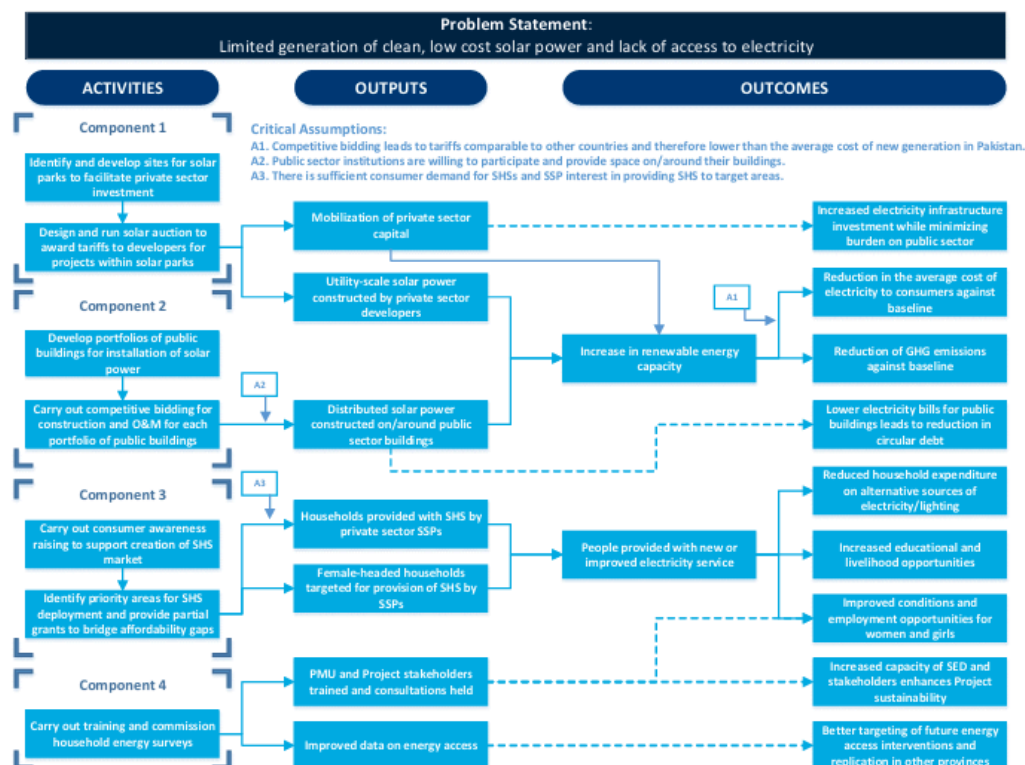
Policy Design

The Sindh Solar Energy Project was designed in response to two related problems: limited access to electricity in parts of Sindh and the wider difficulties facing Pakistan's power sector. Rather than relying on a single approach, the project combined interventions targeting generation, public-sector electricity use, household access, and institutional capacity. Utility-scale solar was intended to increase generation, distributed solar was planned for public-sector buildings, and solar home systems were meant to reach households with limited or no access to electricity. The fourth component focused on capacity building and technical assistance, recognising that implementation would also depend on the ability of government institutions to manage procurement, finance, monitoring and technical requirements.

The Four Components

- Utility-Scale Solar: Development of solar parks and selection of private developers through competitive auctions to build large-scale solar power plants.
- Distributed Solar: Installation of solar PV systems on rooftops and other suitable spaces at public-sector buildings to reduce dependence on conventional electricity.
- Solar Home Systems: Provision of solar home systems to households in areas with limited or no electricity access, along with awareness, financial literacy, quality control, and monitoring.
- Capacity Building & Technical Assistance: Strengthening government and project-management capacity, including procurement, financial management, monitoring and evaluation, gender, safeguards, and technical support.

Theory of Change



Source: World Bank, Sindh Solar Energy Project (P159712): Implementation Completion and Results Report, 9 February 2026,
Figure 1: Theory of Change / Results Chain, p. 3.

Timeline

Key Events	Planned Date	Actual Date
Concept Review	18-Jan-2017	18-Jan-2017
Decision Review	08-Mar-2018	08-Mar-2018
Authorize Negotiations	16-Apr-2018	16-Apr-2018
Approval	14-Jun-2018	14-Jun-2018
Signing	09-Jan-2019	09-Jan-2019
Effectiveness	09-Jan-2019	09-Jan-2019
Restructuring Sequence.01	Not Applicable	24-Jul-2023
ICR Sequence.01 (Final)	--	28-Apr-2026
Mid-Term Review No. 01	29-Apr-2022	11-Mar-2022
Operation Closing/Cancellation	31-Jul-2025	31-Jul-2025
ICR/NCO	30-Apr-2026	29-Apr-2026

Note: World Bank terminology has been clarified for readability. Concept Review and Decision Review refer to internal project review stages; Authorization to Negotiate marks approval to begin financing negotiations; Effectiveness marks the formal start of project implementation; Restructuring refers to a formal modification of the project; Mid-Term Review refers to a review of project progress; ICR means Implementation Completion and Results Report; and NCO means Note on Cancelled Operation.

Design, Risks and Readiness

SSEP was approved in June 2018 with a US\$100 million World Bank credit. The project brought together utility-scale solar, distributed generation, Solar Home Systems and institutional capacity-building. The design therefore required different procurement and implementation arrangements across its components.

The World Bank was fairly explicit about the institutional starting point. Its appraisal noted that the Sindh Energy Department had “no prior experience of implementing multilateral development bank (MDB) projects.” The Bank expected a dedicated Project Management Unit to be established and planned capacity-building in procurement, financial management, monitoring, and safeguards.

Institutional capacity was initially rated Moderate and fiduciary risk Substantial, with a Financial Management Action Plan and simplified procurement strategy built into the project.

Risks also varied by component. Utility-scale solar depended on federal approvals for grid interconnection and power purchase arrangements; distributed solar required coordination among multiple public institutions; and the household component required an effective mechanism to reach priority districts.

Section 2: Implementation of SSEP — What Actually Happened

The SSEP's implementation can be divided into three distinct phases:

Dormancy: 2018 to 2021

The credit became effective on 9 January 2019, seven months after approval; first disbursement in September 2019, fifteen months after approval. A covenant required the project management unit to be appointed within 45 days of effectiveness, and reporting of the later restructuring records a delay of around two years.

Stall: 2022 to mid-2023

The supervision report of May 2023, five months before the original closing date, records no utility-scale capacity built, 439 solar home systems sold against a 200,000 household target, US\$21.27 million disbursed, and both institutional capacity and fiduciary risk escalated to High. The appraisal had planned US\$99.69 million to be disbursed by June 2023. Three shocks fall in this window and are documented by the Bank: the COVID-19 pandemic, the 2022 Sindh floods, and federal import restrictions during Pakistan's balance-of-payments crisis.

Compression: mid-2023 to closure

A restructuring approved on 20 July 2023 extended closure by 22 months to 31 July 2025. Recorded beneficiaries then rose from 1,610 in December 2024 to 321,510 by June 2025. The Senate Standing

Committee was told that the supplier imported 200,968 kits between December 2024 and July 2025, around 30,000 of which could not be distributed within the project period. At closure, the World Bank downgraded the project's headline ratings to Moderately Unsatisfactory, concluding that progress across components was lagging and that completion would require a Government of Sindh-funded Post-Closure Action Plan and continued operation of the PMU.

Performance Against Original Targets

PDO Level

Achievement rate = actual delivery ÷ original target × 100. All "actual" values are the World Bank's own verified end-of-project figures from the final ISR (data date 17 June 2025) unless otherwise stated.

Indicator	Original Target	Deadline	Actual at Close	Achievement	Delay
Renewable generation capacity constructed	420 MW	Dec 2024	31.40 MW	7.5%	Yes
People with new/improved electricity service	1,200,000	Dec 2023	321,510	26.8%	Yes
Households reached (6 persons/household)	200,000	Dec 2023	~53,585	26.8%	Yes

Component Level

Component	Original Target	Budget	Revised Position	Actual Delivery
1. Utility-scale solar	400 MW; US\$273.5m private capital; first 50 MW auction by 31 Dec 2019	\$40m	Budget reduced to ~\$15m; ~270 MW pipeline	0 MW constructed; US\$300 private capital mobilised; auction completed 6 Dec 2024. 0% on capacity.
2. Distributed solar	20 MW on public buildings	\$25m	Raised to ~\$50m; ~50 MW target	31.40 MW constructed. 33 healthcare facilities + 1 water plant completed. 157% vs original target; ~63% vs revised target.
3. Solar home systems	200,000 households; 4,000 female-headed; commercial sale via private providers	\$30m	Shifted to publicly funded free/subsidised distribution via NGOs	~53,585 households (Bank-verified); government figure 126,000 (Jun 2026). 26.8% achievement. Integrity issues identified.
4. Capacity building & TA	Technician training; household surveys; consumer advocate; mid-term audit; 15% female/disabled PMU staffing	\$5m	Majority reallocated to Component 2 (May 2023)	392 of 600 technicians trained (65%); labs established at two universities; consumer advocate underused.

Financial Performance

The financial record is clear evidence of institutional rather than technical failure. Against a plan to disburse US\$99.69 million by June 2023, SSEP had disbursed US\$21.27 million by May 2023, US\$50.44 million by January 2025 and US\$92.60 million at closure, meaning around 42% of the credit moved in the final seven months of a seven-year project. The Bank prints closing disbursement as 98.85%, calculated against the current dollar equivalent of the SDR-denominated credit; against the original US\$100 million it is 92.6%.

More revealing is where the money went. The Auditor-General's audited accounts for 2020-21 and 2021-22, the only years for which component-level statements are public, show Component 1 absorbing 0.9% and then 4.4% of Bank-funded expenditure despite holding 400 of the 420 MW target, while Component 2, holding 20 MW, absorbed 87.7% and then 72.8%; Component 3 received a fifth or less during exactly the years its supplier network was meant to be built, and provincial counterpart funds lapsed unspent in both years. Expenditure tracked procedural friction, not policy priority. The 2021-22 audit opinion was unqualified, which matters for how the later allegations are read: an attest audit tests whether funds were recorded and applied to authorised heads, not whether they bought anything of value, and it long predates the procurement now under investigation.

Outputs Versus Outcomes

Outputs were delivered and verified, whilst the outcomes cannot properly be assessed because the instruments to measure them were not delivered.

The PAD required a household energy survey at the start and end of the project, using the Multi-Tier Framework, as the primary source of outcome and gender-disaggregated data. The endline is not reported in the final ISR results table. There is no published endline dataset.

The social audit required at mid-term (original target December 2020) is recorded in the January 2025 ISR as achieved on 27 September 2024, while the comment on the very same line states: “The Social Audit is delayed and will take place once there is sufficient progress under Component 3 to make it relevant.” The indicator and its own commentary contradict each other.

The consumer advocate was appointed, but the Bank recorded that it was “not clear they are being well utilized.”

The female-headed household indicator is unusable. The final ISR records 20,208 female-headed households provided with SHS against a 4,000 target, but only ≈53,585 households in total, implying a 38% female-headed share against the PAD's stated 1.6% baseline for rural Sindh. The Bank itself flags the indicator as “hard to assess” following the decision to eliminate the differentiated payment for female-headed households. This is a reporting artefact, not a gender outcome, and should not be cited as a success.

Finding: SSEP cannot demonstrate outcomes because it did not build the instruments to measure them. Distribution was verified; functioning, affordability, reliability and welfare change were not. This is a first-order failure independent of the delivery shortfall.

Successful Elements: What Worked and Why

Distributed solar exceeded its target because its delivery chain was short: direct procurement against a defined list of public buildings, with an identifiable public body taking handover, not dependent on private capital, or on tariff determination, or on reaching dispersed rural households. The COVID-19 pivot is the most analytically useful episode in the project: 34 of 35 priority healthcare facilities were contracted and 33 completed under emergency procurement. The same institution, in the same period, delivered quickly once procedural friction was removed. Capacity alone therefore does not explain the wider failure.

The utility-scale component met its demonstration objective while failing its construction objective. Pakistan had not previously run a tariff-based competitive solar auction; the closure document reports

auctions completed at around 3.4 US cents per kilowatt hour with a pipeline of roughly 270 MW, alongside 392 technicians trained and test laboratories established at two universities. Given that the appraisal recorded the implementing agency as having no prior multilateral project experience, this capability transfer is the project's most durable legacy.

Implementation Shortcomings

Utility-scale solar: nothing built. 0 MW against a 400 MW target. Private capital mobilisation was effectively nil against a US\$273.5m target. The cause was sequencing: NEPRA tariff determination, off-taker agreements with CPPA and K-Electric, land allotment not completed until January 2023 and resettlement instruments not disclosed until 2024. None finished early enough for construction to begin. Delivery now depends entirely on provincial follow-through after Bank supervision has ended.

Solar home systems: roughly a quarter delivered. Approximately 53,585 households against 200,000, leaving some 146,000 unreached. Five years of near-inactivity, 439 systems sold by May 2023, were followed by an eight-month procurement sprint. That compression is what created the conditions for the integrity failures below.

Procurement Integrity: Allegations Now Under Investigation

All items below are allegations from unconcluded proceedings and should be attributed, not asserted.

- Sindh cabinet, 1 December 2025: fake import documents, improper contract modifications, payments for incomplete work, distribution partners engaged without open competition, funds spent on undocumented equipment. Cabinet ordered audits, recovery and blacklisting of the main contractor.
- Senate Standing Committee, December 2025 and January 2026: kits quoted to government at US\$112.44 excluding taxes, or US\$151.79 inclusive, against declared customs values of US\$16.1 to US\$23.4; goods declarations later found fake or tampered; US\$12.5m in alleged fake invoices; remittances routed through UAE entities; Anti-Money Laundering Act proceedings opened; no suspensions at that date. Separately, records in Larkana showed seven to eight panels allocated to one family.
- Auditor General of Pakistan, FY2023-24 audit: Rs5.294bn spent with no component complete as at 30 June 2024, and Rs2.182bn of contracts awarded without physical inspection.

Design Drift and Monitoring Failure

Three linked failures leave the project without defensible evidence of what it achieved:

- Component 3 shifted from partial grants to private providers selling commercially, into largely free or subsidised distribution through NGOs, removing the supplier's incentive to maintain a long-term servicing presence and reintroducing the operation and maintenance risk the appraisal document had itself identified.
- The July 2023 restructuring was never reflected in the results framework. The Bank continued reporting Component 2 against the original 20 MW target rather than the revised figure of roughly 50 MW, presenting 157% achievement where the true figure against the operative target is closer to 63%.

- Monitoring instruments were not delivered: no reported endline household survey, a social audit recorded as both complete and delayed, an underused consumer advocate, and internally inconsistent beneficiary indicators.

Undistributed Stock

Around 30,000 procured kits remained undistributed at closure. Public funds were committed and equipment bought without the intended benefit reaching households before Bank oversight ended.

Causes

The single deepest cause is that an implementing agency with no prior MDB experience was given a four-component project with three entirely different delivery models — competitive IPP auctions, public-building EPC, and a consumer-market subsidy scheme — and the risk of this was rated Moderate at appraisal. The Bank's own risk rating for Institutional Capacity moved from Moderate (2018) to High (May 2023), i.e. it was recognised.

A note on external shocks. COVID-19 and the 2022 floods are real and are cited by the Bank and by the government. They should not be allowed to absorb the whole explanation. The PMU covenant was breached before COVID-19; the first 50 MW auction milestone (December 2019) was missed before COVID-19; and Component 2 accelerated during COVID-19 under emergency procurement. External shocks slowed a project that was already not moving.

Equity and Beneficiary Reach

SSEP broadly targeted the right populations, focusing on low-access rural districts and later improving poverty targeting through the BISP/National Socio-Economic Registry. However, delivery integrity was weak, with reports of multiple systems being allocated to single families and NGO distribution occurring without open competition. Gender outcomes also cannot be verified because key monitoring data were missing and the specific support mechanism for female-headed households was removed. Overall, targeting design improved, but there is insufficient evidence that benefits were distributed equitably in practice.

Long-Term Sustainability

SSEP's long-term sustainability remains uncertain because there is no reliable evidence that distributed systems are still functioning or that households have adequate arrangements for repairs, battery replacement and long-term maintenance. The project abandoned the private-sector servicing model originally designed to prevent system failure and instead reverted to a government-led distribution model that the PAD itself had warned was prone to abandonment without sustained O&M. Although the provincial government has retained the PMU and launched a successor solar programme, institutional governance concerns and slow implementation continue to raise doubts about whether benefits will be sustained.

Was There a Compliance Problem?

The Ministry of Economic Affairs asked the World Bank for observations after the September 2025 media reports. Per the project closure document, the World Bank responded that no compliance issues were

identified by its task team or the implementing agency regarding the solar home systems delivered by the contractor.

Two months later, the Sindh cabinet found fake import documents, improper contract modifications and payments for missing work, and ordered blacklisting of the main contractor.

Two months after that, the Senate Standing Committee was briefed on US\$12.5m in alleged fake invoices, tampered goods declarations, and AML proceedings.

This is a direct contradiction between the supervising lender and the borrower's own executive and legislature, on the same facts, within a four-month window.

Section 3: Overall Assessment of the SSEP's Implementation

How Successfully Did SSEP Translate Its Policy Objectives Into Real-World Outcomes?

SSEP was a well-diagnosed project with a theoretically sound method of change, but was largely unsuccessful due to the implementing institution's lack of capacity, and then compromised by the response to that failure.

The project's diagnosis was correct: Sindh had a 37% electricity access rate, excellent solar resource, and no experience of competitive renewable procurement. The design was smart, using public money to de-risk private investment at utility scale, direct procurement where the delivery chain was short, and a market-building subsidy rather than a giveaway at household level. The PAD even identified the exact failure mode that would later occur and designed against it.

What it did not do was match the delivery model to the delivery capability. An agency with no multilateral project experience was tasked with three fundamentally different delivery mechanisms, with institutional capacity risk rated Moderate. The result was five years of near-total inactivity, a 22-month extension, and then a compression of an entire flagship component into eight months. That compression led to fake import documents, threefold price inflation, undistributed stock and an anti-corruption inquiry.

Overall, the SSEP is neither a straightforward failure nor a success. It failed almost completely on physical generation (7.5% of capacity) and substantially on household access (26.8% on the lender's own verified figures). On the other hand, it succeeded in capability transfer. Competitive auctions were run for the first time in Pakistan at 3.4 US cents/kWh, 392 technicians trained, two university test laboratories, a functioning PMU, ~270 MW in a pipeline that did not exist in 2018. It also over-delivered on distributed solar against its original target. It cannot be assessed at all on outcomes, because it did not build the evidence to be assessed by.

The most transferable lesson is that the same institution, in the same period, delivered fastest when procedural friction was removed (the COVID-19 emergency procurement) and failed hardest when it was given a delivery model requiring capabilities it had never had. The binding constraint was institutional design, not solar technology, not finance, and not, despite the pandemic, floods and import restrictions, external shocks alone.

Comparative Research and Recommendations

Sindh's experience with renewable energy shows that having an ambitious solar policy is only the first step. The Sindh Solar Energy Project (SSEP) attempted to expand solar generation and electricity access through utility-scale projects, public buildings, and solar home systems. Looking at renewable-energy policies in other countries shows that the biggest difference between a policy that looks good on paper and one that works in practice often comes down to implementation. Bangladesh, Kenya, and South Africa offer useful examples because each has addressed a different challenge that is also relevant to Sindh: reaching underserved households, encouraging private companies to participate, and making renewable-energy investment more predictable.

Bangladesh

Bangladesh provides one of the strongest comparisons to SSEP because its Solar Home System program faced a similar problem. Many rural households were too far from the conventional grid, making traditional electrification expensive. Instead of having the government purchase and distribute all of the systems itself, Bangladesh created a network through the Infrastructure Development Company Limited (IDCOL), which worked with private companies and NGOs to finance, install, and maintain solar home systems. By 2018, the program had sold more than 4.1 million systems and provided electricity services to around 20 million people.

What makes Bangladesh especially relevant is that it treated solar as a market that needed to be developed, rather than simply as equipment that needed to be distributed. The program included technical standards, consumer financing, quality inspections, and after-sales service. In one World Bank supported program, subsidies were released only after installations were independently verified. SSEP could apply this idea by making part of government payments dependent on whether solar systems remain functional after installation. This would give suppliers a reason to focus on maintenance and reliability instead of treating installation as the end of their responsibility.

Kenya

Kenya offers another useful comparison through its Kenya Off-Grid Solar Access Project. Kenya uses results-based financing to encourage private companies to provide solar systems in remote and underserved areas. Companies receive financial support for achieving specific results, while additional financing helps reduce the high upfront costs associated with entering difficult markets. This approach recognizes that simply telling companies to serve poor or remote communities does not remove the financial risks involved. Sindh could use a similar model by offering incentives for companies that reach underserved communities, but tying those incentives to verified outcomes such as functioning systems, customer service, and maintenance. This would allow the government to use public money to encourage private investment without having to manage every installation directly.

South Africa

South Africa provides a different lesson for SSEP's utility-scale solar component. Its Renewable Energy Independent Power Producer Procurement Programme used competitive tenders to attract private renewable-energy developers. Between 2011 and 2016, the program procured more than 6,300 MW of renewable energy and attracted billions of dollars in private investment. The important lesson is not simply that South Africa used competitive bidding. Its success was also supported by clear requirements, standardized contracts, transparent evaluation, and a relatively predictable process for investors. Sindh could strengthen its own procurement process by making project requirements and timelines clearer and reducing uncertainty between the selection of a developer and the beginning of construction. If private investment is one of the goals of SSEP, then the government should measure whether projects

actually reach financial close and construction, rather than only measuring how much capacity was planned.

Recommendations

These examples suggest that Sindh's main challenge is not a lack of potential. It is creating the systems that allow renewable energy projects to continue working after they are announced or installed. Bangladesh built financing and maintenance into its solar program. Kenya connected government support to measurable results. South Africa created a procurement system that made large renewable-energy investments more attractive to private companies. SSEP included some similar ideas, but future programs could connect them more directly to performance and accountability.

Based on these comparisons, Sindh should focus on three changes:

- Pay for performance rather than simply equipment. Solar systems should be independently verified, and a portion of supplier payments should depend on continued functionality and maintenance.
- Create a stronger private-sector market by establishing clear technical standards, transparent procurement, and financing mechanisms that make it less risky for companies to serve poorer and more remote communities.
- Use data to determine which technology works best in each area. Some communities may be better served by standalone solar systems, while others may eventually need mini-grids or traditional grid expansion.

Sindh should not try to copy another country's policy exactly. The economic and institutional conditions are different. Instead, these examples show that successful renewable-energy policy depends on what happens after a target is announced. The most important shift for Sindh would be to measure success not simply by how many panels are installed or how many megawatts are planned, but by whether those investments continue to provide affordable and reliable electricity. That would make future solar programs more accountable, more attractive to private investors, and more useful to the communities they are intended to serve.

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